

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical

Practical – IX

Roll No.: T001	Name: Soham Acharekar
Class: TYIT	Batch: 01
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Practical 9: Hypothesis Testing and A/B Testing

Part A: Hypothesis Testing

Dataset: supermarket_sales.csv

1. Load the supermarket_sales.csv dataset.
2. Display the first five records.
3. Select the revenue column.
4. Check for missing values.
5. Calculate the mean revenue.
6. Assume that the expected average revenue is ₹1,500. 7. Write the null hypothesis:

$$H_0: \mu = 1500$$

8. Write the alternative hypothesis:

$$H_1: \mu \neq 1500$$

9. Set the significance level as 5%.
10. Perform a one-sample t-test.
11. Calculate the p-value.
12. Compare the p-value with 0.05.
13. State whether the null hypothesis is rejected or not rejected. 14. Write the conclusion in simple business terms.

CODE:

```
import pandas as pd
import scipy.stats as stats
```

```

# Load dataset
df = pd.read_csv("supermarket_sales.csv")
# Show first 5 records
print(df.head())
# Use 'revenue' column for hypothesis testing
revenue = df['revenue']
# Check missing values
print("Missing values:", revenue.isnull().sum())
# Mean revenue
mean_revenue = revenue.mean()
print("Mean Revenue:", mean_revenue)
# Hypothesis setup
expected_mean = 1500
alpha = 0.05
# One-sample t-test
t_stat, p_value = stats.ttest_1samp(revenue, expected_mean)
print("T-statistic:", t_stat)
print("P-value:", p_value)
# Decision
if p_value < alpha:
    print("Reject Null Hypothesis")
    print("Conclusion: The average revenue is significantly different
from ₹1500.")
else:
    print("Fail to Reject Null Hypothesis")
    print("Conclusion: The average revenue is not significantly
different from ₹1500.")

```

OUTPUT:

```

...      invoice_id branch      city customer_type gender_customer \
0    750-67-8428      A    Yangon      Member      Female
1    226-31-3081      C  Naypyitaw      Normal      Female
2    631-41-3108      A    Yangon      Normal      Male
3    123-19-1176      A    Yangon      Member      Male
4    373-73-7910      A    Yangon      Normal      Male

      product_line  unit_cost  quantity  Spct_markup  revenue \
0    Health and beauty      74.69         7      26.1415  548.9715
1  Electronic accessories      15.28         5       3.8200   80.2200
2    Home and lifestyle      46.33         7      16.2155  340.5255
3    Health and beauty      58.22         8      23.2880  489.0480
4    Sports and travel      86.31         7      30.2085  634.3785

      date   time payment_method  cogs  gm_pct  gross_income  rating
0  01/05/19  13:08      Ewallet  522.83  4.761905      26.1415    9.1
1  03/08/19  10:29        Cash   76.40  4.761905       3.8200    9.6
2  03/03/19  13:23  Credit card  324.31  4.761905      16.2155    7.4
3  1/27/2019  20:33      Ewallet  465.76  4.761905      23.2880    8.4
4  02/08/19  10:37      Ewallet  604.17  4.761905      30.2085    5.3
Missing values: 0
Mean Revenue: 322.966749
T-statistic: -151.37567896776733
P-value: 0.0
Reject Null Hypothesis
Conclusion: The average revenue is significantly different from ₹1500.

```

Part B: A/B Testing**Dataset: ab_testing.csv**

The dataset contains:

Group	Users	Conversions
A	1000	120
B	1000	150

1. Load the A/B testing dataset.
2. Display the data.
3. Calculate the conversion rate for Group A.
4. Calculate the conversion rate for Group B.
5. Compare the conversion rates of A and B.
6. Write the null hypothesis:
 7. $H_0: p_A = p_B$
8. Write the alternative hypothesis:
 9. $H_1: p_A \neq p_B$
10. Set the significance level as 5%.
11. Perform a two-proportion test.
12. Calculate the p-value.
13. Compare the p-value with 0.05.
14. State whether the null hypothesis is rejected or not rejected. 13. Create a bar chart to compare the conversion rates. 14. Write the conclusion based on the test result.

CODE:

```
import numpy as np
from statsmodels.stats.proportion import proportions_ztest
import matplotlib.pyplot as plt
# A/B testing dataset
data = {
    'Group': ['A', 'B'],
    'Users': [1000, 1000],
    'Conversions': [120, 150]
}
# Display data
print(data)
# Conversion rates
conv_rate_A = data['Conversions'][0] / data['Users'][0]
conv_rate_B = data['Conversions'][1] / data['Users'][1]
print("Conversion Rate A:", conv_rate_A)
print("Conversion Rate B:", conv_rate_B)
# Hypothesis setup
alpha = 0.05
# Two-proportion z-test
```

```

successes = np.array([data['Conversions'][0], data['Conversions'][1]])
nobs = np.array([data['Users'][0], data['Users'][1]])
z_stat, p_value = proportions_ztest(successes, nobs, alternative='two-
sided')
print("Z-statistic:", z_stat)
print("P-value:", p_value)
# Decision
if p_value < alpha:
    print("Reject Null Hypothesis")
    print("Conclusion: Conversion rates between Group A and Group B are
significantly different.")
else:
    print("Fail to Reject Null Hypothesis")
    print("Conclusion: No significant difference in conversion rates
between Group A and Group B.")
# Bar chart
plt.bar(data['Group'], [conv_rate_A, conv_rate_B], color=['blue', 'green'])
plt.ylabel("Conversion Rate")
plt.title("A/B Testing Conversion Rates")
plt.show()

```

OUTPUT:

```

...  {'Group': ['A', 'B'], 'Users': [1000, 1000], 'Conversions': [120, 150]}
      Conversion Rate A: 0.12
      Conversion Rate B: 0.15
      Z-statistic: -1.963049807622355
      P-value: 0.04964038686536878
      Reject Null Hypothesis
      Conclusion: Conversion rates between Group A and Group B are significantly different.

```

